A Proof of The First-Token Domination

Proof. Without loss of generality, we assume the LM have L layers with only 1 attention head ($h = H = 1$), and the attention weight matrix in layer l is $\boldsymbol{A}(l,h)$. We consider the input that have more than 1 token. We model attention as flows following the setting of Abnar and Zuidema (2020). Then, the attention accumulation Accum() of the token $\boldsymbol{z}_i^{l+1}$ in the layer $l+1$ can be written as

$$\text{Accum}(\boldsymbol{z}_i^{l+1}) = \sum_{1 \le j \le |T|} \boldsymbol{a}_{i,j}(l,h) \cdot \text{Accum}(\boldsymbol{z}_j^l), \tag{3}$$

where we have

$$\text{Accum}(\boldsymbol{z}_i^1) = \sum_{1 \le j \le |T|} \boldsymbol{a}_{i,j}(l,h) \cdot t_j. \tag{4}$$

Since $\boldsymbol{a}_{i,j}(l,h)$ is the attention weight for casual LM, we have

$$\begin{cases} \boldsymbol{a}_{i,j}(l,h) = 0 & \text{if} \quad i < j \\ 0 < \boldsymbol{a}_{i,j}(l,h) < 1 & \text{if} \quad i \ge j \\ \sum_j \boldsymbol{a}_{i,j}(l,h) = 1 \end{cases}. \tag{5}$$

Note that attention is normalized by *Softmax* function in LMs. The minimum attention weight is non-zero, and we assume there exist a constant $\epsilon > 0$ such as $\epsilon \le \boldsymbol{a}_{i,j}(l,h) < 1$ if $i \ge j$. Now we define the information ratio $\text{IR}_{\boldsymbol{z}_i^l}(t_j)$ as the information of token t_j stored in the hidden representation $\boldsymbol{z}_i^l$. Consider that in each layer, token t_1 would propagate its information to all tokens in the next layer with at least ϵ amount. Then, by tracing the information flow from other tokens $j > 1$, we have

$$1 - \text{IR}_{\boldsymbol{z}_i^{l+1}}(t_1) \le (1-\epsilon)(1 - \text{IR}_{\boldsymbol{z}_i^l}(t_1)). \tag{6}$$

Using the chain rule, we have

$$1 - \text{IR}_{\boldsymbol{z}_i^L}(t_1) \le (1-\epsilon)^L (1 - \text{IR}_{\boldsymbol{z}_i^1}(t_1)), \tag{7}$$

which means

$$\text{IR}_{\boldsymbol{z}_i^L}(t_1) \ge 1 - (1-\epsilon)^{L-1}(1 - \text{IR}_{\boldsymbol{z}_i^1}(t_1)), \tag{8}$$

$$\text{IR}_{\boldsymbol{z}_i^L}(t_1) \ge 1 - (1-\epsilon)^L. \tag{9}$$

With the inequality above, we know that if the LM is deep, i.e., L is large, we have $\text{IR}_{\boldsymbol{z}_i^L}(t_1)$ increase exponentially in terms of layer L, which means that $\text{IR}_{\boldsymbol{z}_i^L}(t_1) \approx 1$ with large L in general. □

B Supplementary about GPT-2

We provide some more details on our experiments on GPT-2 as well as LLaMA to help in reproducibility. First of all, we fix the random seed (42) and use the same random seed for all experiments, including LM finetuning and interpretations. In addition, to make sure the random seed is unbiased, we further re-run the same experiment with different random seeds. All of our experiments have roughly the same results as those of using other seeds.

Second, we design our analysis as simply as possible to ensure that there is as little random influence (i.e., confounder) as possible. For `MechanisticProbe`, we select the kNN classifier. For LLaMA, we run analysis experiments on a 4-shot in-context learning setting.

Third, we report as many intermediate and supplement results as possible. In the Appendix, there are many other interesting findings. However, due to the space limit, we cannot present them in the main paper. We hope our findings are helpful to the community to better understand LMs.

B.1 GPT-2 Finetuning

The finetuning settings for all GPT-2 models are roughly identical. We generate 0.98 million sequences of numbers as the training data and $10,000$ in data for validation and testing. Note that the collision probability is extremely small, thus we can assume that there is no data leakage. The epoch number is set as 2, and the batch size here is 256. We use the AdamW (Loshchilov and Hutter, 2019) optimizer with weight decay $1e-3$ from Huggingface[13] for finetuning. The learning rate is $1e-6$.

B.2 Original Probing Scores

The original classification scores (F1-Macro) for two probing tasks can be found in Table 4. Here, *random* means we randomly initialize GPT-2 and use its attentions for probing as the random baseline. The other two *pretrained* and *finetuned* are the pretrained GPT-2 model and GPT-2 model after finetuning with supervised signals.

B.3 Visualization of $\mathbb{E}[\pi(\boldsymbol{A}_{\text{simp}})]$ for GPT-2

We visualize the $\boldsymbol{A}_{\text{simp}}$ for pretrained GPT-2 without finetuning in Figure 11. We can find that even if pretrained GPT-2 cannot solve the synthetic reasoning task (finding k-th smallest number from a list). It can still somehow differentiate the size of

[13] https://huggingface.co/

Table 4: The original classification F1-Macro scores of GPT-2 for two probing tasks.

Probing task	$S_{\text{F1}}(V\|\boldsymbol{A}_{\text{simp}})$			$S_{\text{F1}}(G\|V,\boldsymbol{A}_{\text{simp}})$		
GPT-2	random	pretrained	finetuned	random	pretrained	finetuned
$k=1$	48.38	52.04	96.36	100	100	100
$k=2$	48.60	51.61	96.77	73.18	78.72	99.47
$k=3$	47.62	54.68	95.61	61.45	66.69	98.36
$k=4$	47.42	58.32	93.87	55.71	59.28	96.69
$k=5$	48.50	60.94	93.58	54.86	55.48	94.66
$k=6$	48.66	61.40	93.27	50.44	55.98	97.55
$k=7$	49.28	62.40	88.14	51.75	51.11	97.55
$k=8$	49.74	60.95	89.31	50.54	51.07	97.00

numbers. The largest number of the list often has slightly larger attentions in layer 11.

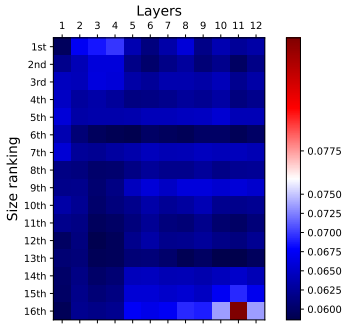


Figure 11: Visualization of $\mathbb{E}[\pi(\boldsymbol{A}_{\text{simp}})]$ for pretrained GPT-2 (w/o any finetuning). Similarly, we take mean pooling for different attention heads.

B.4 Visulization of $\boldsymbol{A}$

To avoid disturbance, we remove 40% position heads (heads with small position entropy) and take mean pooling on all left heads. We visualize $\boldsymbol{A}$ to directly show that $\boldsymbol{A}_{\text{simp}}$ contains sufficient essential information of $\boldsymbol{A}$. We consider a special case when $k=2$, the largest number is at position 8 and the second largest number is at position 12.[14]

The attention $\mathbb{E}[\boldsymbol{A}]$ is visualized as in Figure 12. From Figure 12, we can get similar conclusion as on the visualization of $\mathbb{E}[\boldsymbol{A}_{\text{simp}}]$. In the bottom layers, most hidden representations focus on top-2 numbers. In the top layers, most hidden representations focus on the second smallest number. This result proves that our analysis on $\boldsymbol{A}_{\text{simp}}$ is as reasonable as that on $\boldsymbol{A}$.

We also provide the visualization of $\mathbb{E}[\boldsymbol{A}]$ on normal test data (i.e., the input position is independent of the size) in Figure 13. We can find that the attention distribution is even, and there is no typical tendency.

[14]We modify our test data only to satisfy this condition.

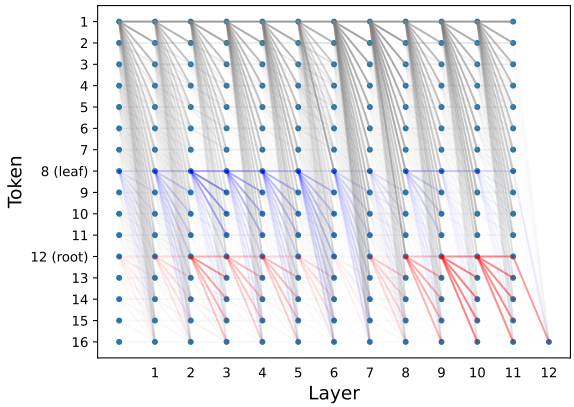


Figure 12: Visualization of $\mathbb{E}[\boldsymbol{A}]$. The x-axis represents layers and the y-axis represents input token positions. We take the mean pooling of 60% attention heads with large position entropy. The leaf node (i.e., the largest number) is always at position 8, and the root node (i.e., the second largest number) is always at position 12. We track their attentions and visualize them by blue and red lines.

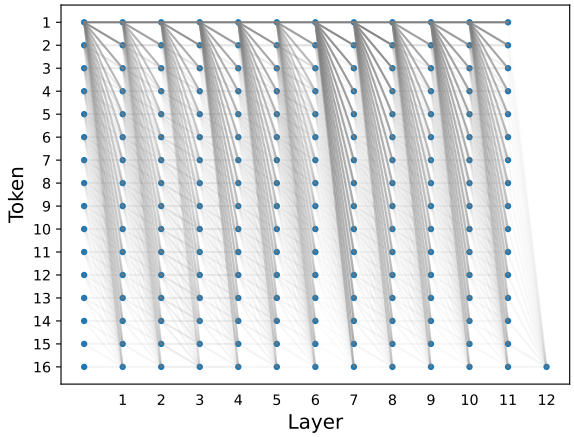


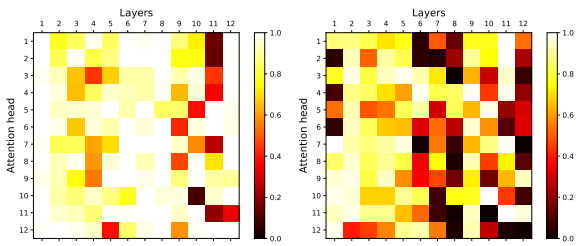
Figure 13: Visualization of $\mathbb{E}[\boldsymbol{A}]$ on normal test data. The x-axis represents layers and the y-axis represents input token positions. We take mean pooling of 60% attention heads with large position entropy.

B.5 Attention Head Entropy

From Figure 14, we can find that most heads belong to either position head or size head. Note that we generate input data randomly, thus, the number size and input position are independent of each other. Thus, one head cannot be both position head and size head.

B.6 Do Finetuning Methods Matter?

To show that our analysis is robust, we explore the attention of GPT-2 with different ways of param-



(a) Entropy in terms of number size (rank)

(b) Entropy in terms of number input position

Figure 14: Entropy for all attention heads of finetuned GPT-2. x-axis represents layers and y-axis represents the head index. The cube (better view in color) shows the entropy value.

eter efficient finetuning (Ruder et al., 2022). We report our two probing scores S_{P1} and S_{P2} with various ways of finetuning in Figure 15 (We consider the condition when $k = 2$ and $m = 16$). We find that probing scores of finetuning the full GPT-2 model are similar to that of partially finetuning on attention parameters and MLP (multilayer perceptron) parameters. These consistent results ensure the general usage of our `MechanisticProbe` on current large LMs with partial finetuning.

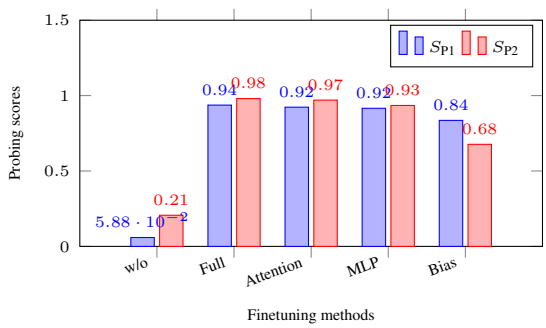
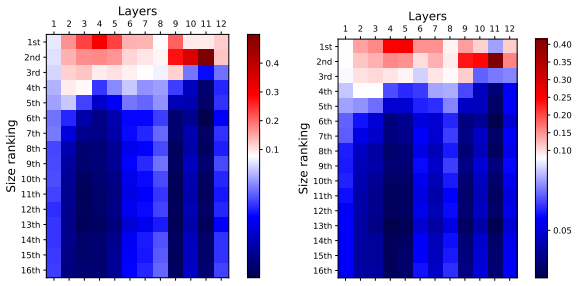


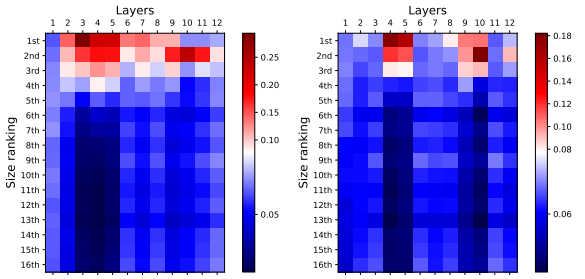
Figure 15: Probing scores of GPT-2 with different finetuning methods. Here, *w/o* denotes the baseline when GPT-2 model is not finetuned. *Full* denotes finetuning with all parameters. Other models are partial finetuned with corresponding parameters. With the exception of probing scores of the *Bias* tuned model, other partial finetuning methods have roughly the same probing scores compared to that full finetuning. This indicates that `MechanisticProbe` can provide consistent analysis for LMs finetuned in different ways.

The direct visualization of $\mathbb{E}[\pi(\boldsymbol{A}_{\text{simp}})]$ for different finetuning methods can be found Figure 16(a-d). And the test accuracy of these 4 finetuned models are: 99.42, 99.31, 94.11, and 76.84. We can find that finetuning attention parameters or MLP parameters can obtain quite similar attention patterns to that of full model finetuning. Regarding the bias tuning, it is slightly different. We speculate that this is because the LM does not learn the k-th smallest element task well (with much lower test accuracy). Generally, without significant performance drops, these results can support the general usage of our probing method `MechanisticProbe` under different finetuning settings.



(a) Full

(b) Partial (Attention)



(c) Partial (MLP)

(d) Partial (Bias)

Figure 16: Visualization of $\mathbb{E}[\pi(\boldsymbol{A}_{\text{simp}})]$ with different ways of finetuning. The x-axis represents layers and the y-axis represents size ranking. We take mean pooling attention heads. From Figures (a-d) are GPT-2 with full finetuning, GPT-2 with partial finetuning on attention parameters, GPT-2 with partial finetuning on MLP parameters, and GPT-2 with partial finetuning on bias parameters. We can find that different finetuning methods would not affect the attention patterns much.

B.7 Task Difficulty & Model Capacity

This subsection discusses which factors can let GPT-2 handle reasoning tasks with more leaf nodes in G, i.e., large k. We explore the model capacity and reasoning task difficulty. Specifically, we extend the list length m from 16 to 64, and maximum k from 8 to 32. For each k, we finetune LMs with the same finetuning settings and evaluate them on test data with accuracy. For model capacity, we compare three versions of GPT-2 with different sizes: Distilled GPT-2, GPT-2 (small), and GPT-2 Medium. For task difficulty, we construct synthetic data by selecting $m = 64$ numbers from 256, 384,